Experiment 1 : Data Loading, Storage and File Form

Aim: To load and analyze sales data from multiple file formats (CSV, Excel and JSON), and perf cleaning, transformation and analysis on the dataset.

In [1]:
```
# A52 Dipanshu Ambilkar
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

In [2]:
```
np.random.seed(1)
n = 120
df = pd.DataFrame({
    "OrderID": range(1, n + 1),
    "Product": np.random.choice(["Laptop", "Phone", "Tablet", "Headphones
    "Category": np.random.choice(["Electronics", "Accessories"], n),
    "Quantity": np.random.randint(1, 5, n),
    "Price": np.random.choice([999, 1499, 2499, 4999, 12999], n),
    "Region": np.random.choice(["North", "South", "East", "West"], n),
})
df.head()
```

Out[2]:

	OrderID	Product	Category	Quantity	Price	Region
0	1	Phone	Electronics	2	1 4 9 9	South
1	2	Headphones	Accessories	1	2 4 9 9	South
2	3	Laptop	Accessories	2	1 4 9 9	South
3	4	Laptop	Accessories	1	9 9 9	North
4	5	Headphones	Accessories	2	9 9 9	East

In [3]:
```
df.loc[np.random.choice(n, 8), "Price"] = np.nan
df = pd.concat([df, df.iloc[:5]], ignore_index=True)
print("shape after adding missing values and duplicates:", df.shape)
```

```
shape after adding missing values and duplicates: (125, 6)
```

In [4]:
```
df.to_csv("sales.csv", index=False)
df.to_excel("sales.xlsx", index=False)
df.to_json("sales.json", orient="records")
print("saved sales.csv, sales.xlsx, sales.json")
```

```
saved sales.csv, sales.xlsx, sales.json
```

In [5]:
```
c = pd.read_csv("sales.csv")
e = pd.read_excel("sales.xlsx")
j = pd.read_json("sales.json")
print("shapes:", c.shape, e.shape, j.shape)
```

```
shapes: (125, 6) (125, 6) (125, 6)
```

In [6]:
```
print("missing values in csv data:")
c.isnull().sum()
```

```
missing values in csv data:
```

Out[6]:
```
OrderID     0
Product     0
Category    0
Quantity    0
Price       8
Region      0
dtype: int64
```

In [7]:
```
c = c.drop_duplicates()
c["Price"] = c["Price"].fillna(c["Price"].median())
print("rows in csv data now:", len(c))
print("missing prices now:", c["Price"].isnull().sum())
```

```
rows in csv data now: 120
missing prices now: 0
```

In [8]:
```
e = e.drop_duplicates()
e["Price"] = e["Price"].fillna(e["Price"].median())
print("rows in excel data now:", len(e))
print("missing prices now:", e["Price"].isnull().sum())
```

```
rows in excel data now: 120
missing prices now: 0
```

In [9]:
```
c["Total"] = c["Quantity"] * c["Price"]
c.head()
```

Out[9]:

	OrderID	Product	Category	Quantity	Price	Region	Total
0	1	Phone	Electronics	2	1499.0	South	2998.0
1	2	Headphones	Accessories	1	2499.0	South	2499.0
2	3	Laptop	Accessories	2	1499.0	South	2998.0
3	4	Laptop	Accessories	1	999.0	North	999.0
4	5	Headphones	Accessories	2	999.0	East	1998.0

In [10]:
```
print(c.describe())
```

```
         OrderID    Quantity         Price         Total
count  120.000000  120.000000    120.000000    120.000000
mean    60.500000    2.500000   4778.166667  11180.833333
std     34.785054    1.173947   4610.362749  12336.461920
min      1.000000    1.000000    999.000000    999.000000
25%     30.750000    1.000000   1499.000000   2997.000000
50%     60.500000    3.000000   2499.000000   5996.000000
75%     90.250000    4.000000   4999.000000  12999.000000
max    120.000000    4.000000  12999.000000  51996.000000
```

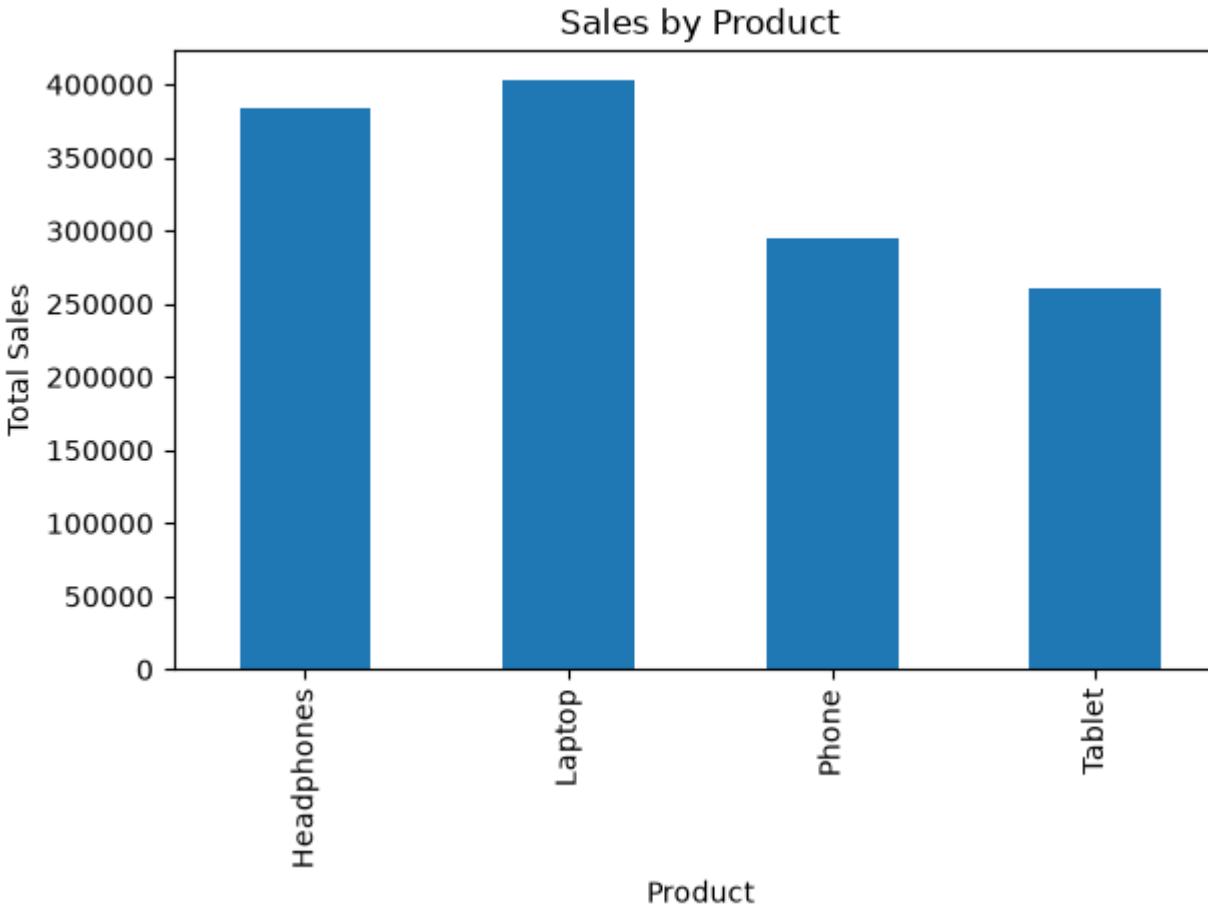
In [11]:
```
print("Total sales:", c["Total"].sum())
print("Avg order value:", c["Total"].mean())
```

```
Total sales: 1341700.0
Avg order value: 11180.833333333334
```

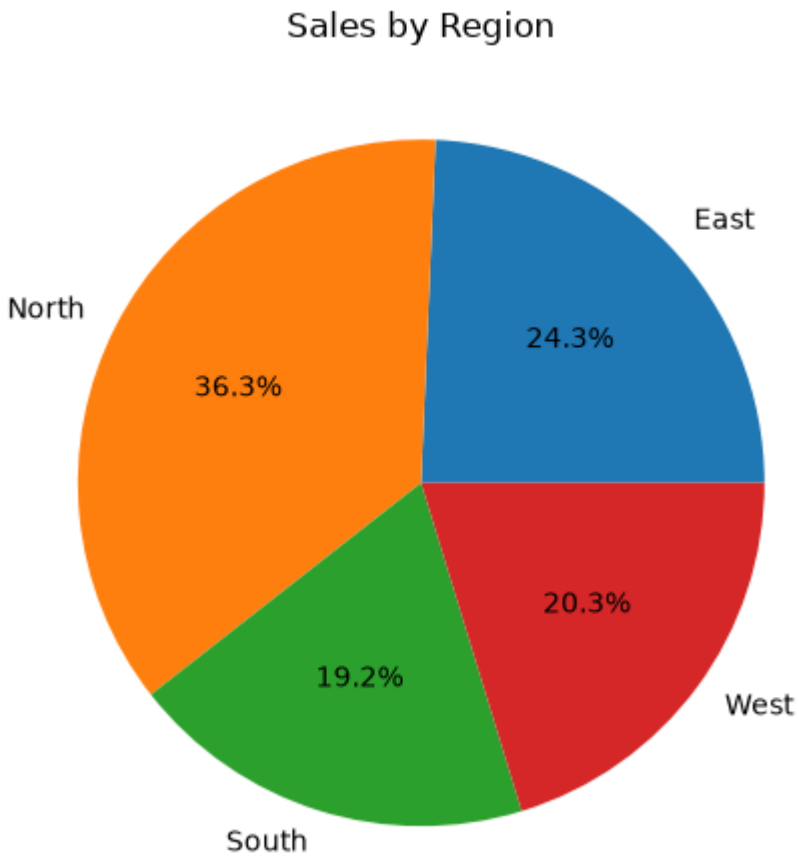
In [12]:
```
c.groupby("Category")["Total"].sum()
```

Out[12]:
```
Category
Accessories    693855.0
Electronics    647845.0
Name: Total, dtype: float64
```

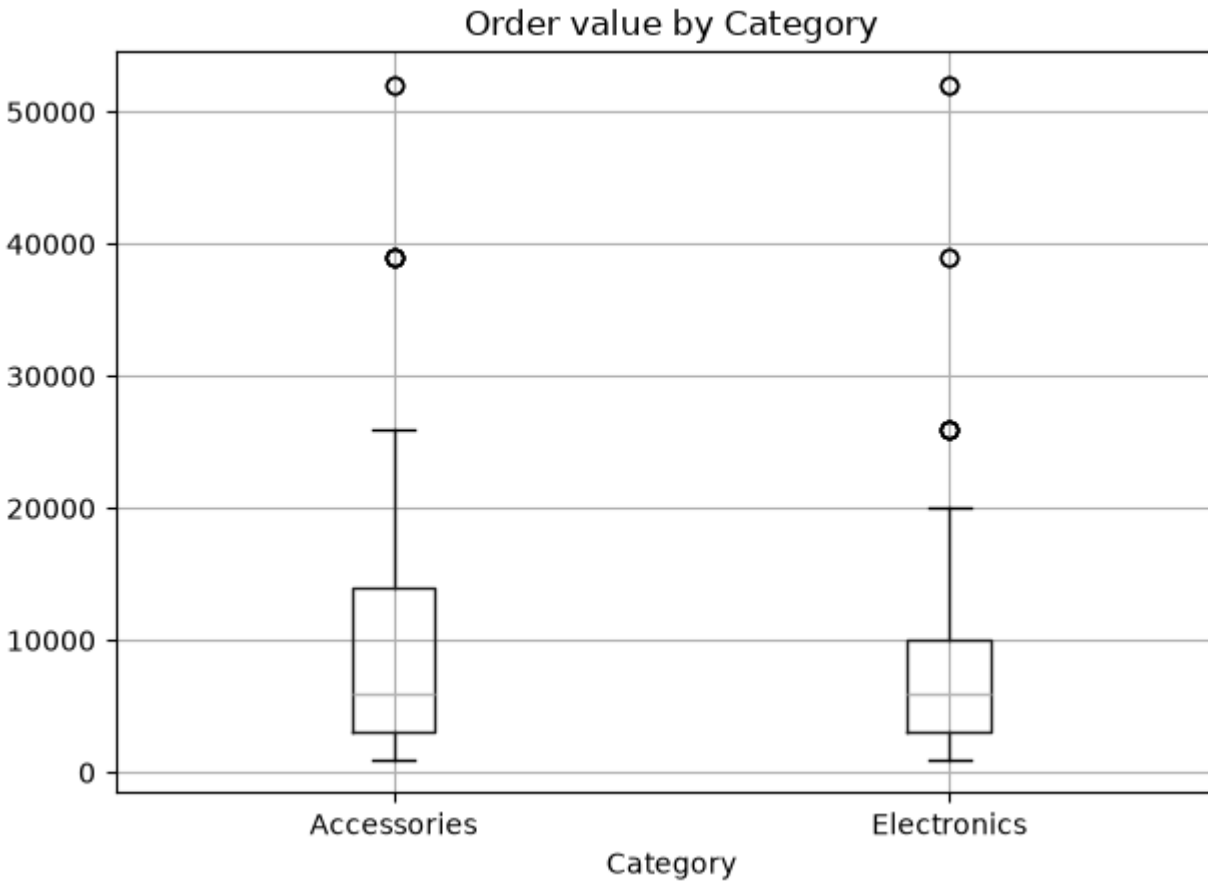
In [13]:
```
c.groupby("Product")["Total"].sum().plot(kind="bar")
plt.title("Sales by Product")
plt.ylabel("Total Sales")
plt.tight_layout()
plt.show()
```



In [14]:
```
c.groupby("Region")["Total"].sum().plot(kind="pie", autopct="%1.1f%%")
plt.title("Sales by Region")
plt.ylabel("")
plt.tight_layout()
plt.show()
```



In [15]:
```
c.boxplot(column="Total", by="Category")
plt.title("Order value by Category")
plt.suptitle("")
plt.tight_layout()
plt.show()
```



In [16]:
```
print("done")
```

```
done
```